End-sem: worked solutions

25 Nov 2024 · 40 marks · 3 hours · CEPC12

Question 1

Q1 (a) 2 marks

> Explain the relationship between absolute, gauge and vacuum pressure. Find the gauge and absolute pressure at a point 2.0 m below the free surface of water. Atmospheric pressure = 10.1043 N/cm².

- **Absolute pressure** is measured from absolute zero (perfect vacuum).
- **Gauge pressure** is measured from local atmospheric pressure: $p_{abs} = p_{atm} + p_{gauge}$.
- **Vacuum (negative gauge) pressure** is the amount by which pressure is below atmospheric: $p_{abs} = p_{atm} - p_{vac}$.

$p_{gauge} = \rho g h = 9810 \times 2 = 19{,}620\ \text{N/m}^2 =$ **1.962 N/cm²**

$p_{abs} = 10.1043 + 1.962 =$ **12.066 N/cm²**

Q1 (b) 2 marks

> Define surface tension. Prove that the excess pressure inside a liquid droplet is p = 4σ/d. The pressure outside a 0.04 mm water droplet is 10.32 N/cm². Find the pressure inside if σ = 0.0725 N/m.

Surface tension (σ): the tensile force per unit length acting along the free surface of a liquid, due to cohesion between surface molecules (N/m).

Proof: cut the droplet of diameter d into two halves. On one half:

- pressure force on the projected area = $p \times \pi d^2/4$ (p = excess pressure inside)

- surface tension force around the circumference = $\sigma \times \pi d$

Equilibrium: $p \times \pi d^2/4 = \sigma \times \pi d \implies$ **$p = 4\sigma/d$**

$p = 4 \times 0.0725 / (0.04 \times 10^{-3}) = 7250\ N/m^2 = 0.725\ N/cm^2$

Pressure inside = 10.32 + 0.725 = **$11.045\ N/cm^2$**

Q1 (c)

3 marks

> Define and explain Newton's law of viscosity. The space between two square flat parallel plates (side 60 cm) is filled with oil 12.5 mm thick. The upper plate moves at 2.5 m/s and needs a force of 98.1 N. Find (i) the dynamic viscosity in poise, (ii) the kinematic viscosity in stokes if S = 0.95.

Newton's law of viscosity: the shear stress on a layer of fluid is **directly proportional to the rate of shear strain (velocity gradient)**: $\tau = \mu\, du/dy$. Fluids that obey it (water, air, oil) are **Newtonian**; μ is the constant of proportionality (dynamic viscosity).

$A = 0.6 \times 0.6 = 0.36\ m^2$; $\tau = 98.1/0.36 = 272.5\ N/m^2$

(i) $\mu = \tau\, dy/du = 272.5 \times 0.0125/2.5 = 1.3625\ N\cdot s/m^2 =$ **13.63 poise**

(ii) $\nu = \mu/\rho = 1.3625/950 = 1.434 \times 10^{-3}\ m^2/s =$ **14.34 stokes**

Question 2

Q2 (a)

2 marks

> Explain Pascal's law. A 6 m deep tank contains 4 m of water and 2 m of oil (RD 0.88). Find the pressure at the bottom.

Pascal's law: the pressure at a point in a fluid at rest is the **same in all directions** (and a pressure applied to a confined fluid is transmitted equally throughout it).

$p = \rho_{oil}\, g\, h_{oil} + \rho_w\, g\, h_w = 0.88 \times 9810 \times 2 + 9810 \times 4 = 17{,}266 + 39{,}240$

$p =$ **$56{,}506\ N/m^2 \approx 56.5\ kPa$ (gauge)**

Q2 (b)

3 marks

> Differentiate between simple and differential manometers. The figure shows an inverted differential manometer connected to pipes A and B containing water; the manometer fluid is oil of S = 0.8. Find the difference of pressure head between A and B.

Simple manometer	Differential manometer
Measures pressure at **one point** relative to atmosphere	Measures the **difference** in pressure between **two points**
One end connected to the pipe, the other open to atmosphere	Both ends connected to pipes/points
Piezometer, U-tube, single-column	U-tube differential, inverted U-tube differential

Reading the figure: the oil–water interface in the left limb is 100 cm above A; the interface in the right limb is 30 cm lower; B is 50 cm below the right interface.

Left interface: $p = p_A - 9810 \times 1.0$

Right interface (0.3 m of oil lower): $p = p_A - 9810 \times 1.0 + 800 \times 9.81 \times 0.3$

Also, right interface = $p_B - 9810 \times 0.5$

$p_A - p_B = 9810 - 4905 - 2354.4 = 2550.6\ \text{N/m}^2$

Difference of pressure head = 2550.6/9810 = **0.26 m of water** (A higher)

Q2 (c)

3 marks

> A dam has a parabolic face $y = y_0(x/x_0)^2$ with $x_0 = 6$ m and $y_0 = 9$ m. Water (1000 kg/m^3) stands to the top. Find the horizontal, vertical and resultant thrust per metre length.

Horizontal thrust = force on the vertical projection (9 m high, 1 m wide):

$F_H = \rho g\,(H/2)\,H = 9810 \times 4.5 \times 9 =$ **397.3 kN/m**, acting $H/3 = 3$ m above the base

Vertical thrust = weight of water above the curved face. The face is $x = x_0\sqrt{(y/y_0)} = 2\sqrt{y}$, and the water lies between this curve and the y-axis:

Area $= \int_0^9 x\,dy = \int_0^9 2\sqrt{y}\,dy = (4/3)\,y^{3/2}\,|_0^9 = 36\ m^2$

$F_V = 9810 \times 36 \times 1 =$ **353.2 kN/m**, acting at $\bar{x} = \int x(9 - y)dx / 36 = 81/36 = 2.25$ m from O

Resultant:

$F = \sqrt{(397.3^2 + 353.2^2)} =$ **531.6 kN/m**, at $\theta = \tan^{-1}(353.2/397.3) = 41.6°$ to the horizontal

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Q2 (d)

3 marks

> Differentiate between local and convective acceleration. The stream function of a 2-D flow is $\psi = 2xy$. Find the velocity at P(2, 3) and the velocity potential ϕ.

- **Local (temporal) acceleration**: the rate of change of velocity **with time at a fixed point**, $\partial u/\partial t$. Present only in **unsteady** flow.
- **Convective acceleration**: the rate of change of velocity **due to change of position** in the flow field, $u\,\partial u/\partial x + v\,\partial u/\partial y + w\,\partial u/\partial z$. Present in **non-uniform** flow (e.g. a tapering pipe), even if steady.

$u = \partial\psi/\partial y = 2x = 4$; $v = -\partial\psi/\partial x = -2y = -6$ at P(2, 3)

Resultant velocity $= \sqrt{(4^2 + 6^2)} =$ **7.21 units**, at $\tan^{-1}(6/4) = 56.3°$ below the x-axis

$\partial\phi/\partial x = u = 2x \rightarrow \phi = x^2 + f(y)$; $\partial\phi/\partial y = f'(y) = -2y \rightarrow$ **$\phi = x^2 - y^2 + C$**

Question 3

Q3 (a)

3 marks

> What is Couette flow? Water at 15 °C flows between two large parallel plates 1.6 mm apart. Find (i) the maximum velocity, (ii) the pressure drop per unit length and (iii) the wall shear stress, if the average velocity is 0.2 m/s. $\mu = 0.01$ poise.

Couette flow: laminar flow between two parallel plates where **one plate moves** relative to the other; the flow is driven by the moving plate (shear) with or without a pressure gradient. With no pressure gradient the velocity profile is **linear**, $u = U\,y/t$.

(The numerical is plane Poiseuille flow between **fixed** plates.) $\mu = 0.01$ poise $= 0.001$ N·s/m², $t = 1.6$ mm.

(i) $u_{max} = 1.5\,\bar{u} = 1.5 \times 0.2 =$ **0.3 m/s**

(ii) $-dp/dx = 12\mu\bar{u}/t^2 = 12 \times 0.001 \times 0.2 / (0.0016)^2 =$ **937.5 N/m² per m**

(iii) $\tau_0 = (-dp/dx)\,t/2 = 937.5 \times 0.0008 =$ **0.75 N/m²**

Q3 (b)

3 marks

> Three pipes of 400, 200 and 300 mm diameter have lengths 400, 200 and 300 m and are connected in series. The ends are connected to two tanks with a level difference of 16 m. $f = 0.005$ for all. Find the discharge, first neglecting and then including minor losses.

Using $h_f = 4fLV^2/(2gd)$ and writing everything in terms of V_2 (200 mm pipe): $V_1 = V_2/4$, $V_3 = (2/3)^2 V_2 = 0.444V_2$.

$\Sigma\, LV^2/d = 400(V_2/4)^2/0.4 + 200V_2^2/0.2 + 300(0.444V_2)^2/0.3 = (62.5 + 1000 + 197.5)V_2^2 = 1260.0\,V_2^2$

Friction loss $= (4 \times 0.005/19.62) \times 1260.0\,V_2^2 = 1.2844\,V_2^2$

Neglecting minor losses: $16 = 1.2844\,V_2^2 \rightarrow V_2 = 3.529$ m/s

$Q = \pi/4 \times 0.2^2 \times 3.529 =$ **0.1109 m³/s**

Minor losses ($\times V_2^2/2g$): entrance $0.5(1/16) = 0.031$; contraction 0.5; enlargement $(1 - 0.444)^2 = 0.309$; exit $0.444^2 = 0.198$; total $1.037\,V_2^2/2g = 0.0529\,V_2^2$.

Including minor losses: $16 = (1.2844 + 0.0529)V_2^2 \rightarrow V_2 = 3.459$ m/s

$Q = \pi/4 \times 0.2^2 \times 3.459 =$ **0.1087 m³/s**

Q3 (c)

2 marks

> Water flows over a rectangular weir 1 m wide at a depth of 150 mm and then through a right-angled triangular weir. $C_d = 0.62$ (rectangular) and 0.59 (triangular). Find the depth over the triangular weir.

$Q = (2/3)\, C_d\, L\, \sqrt{(2g)}\, H^{3/2} = (2/3)(0.62)(1)(4.429)(0.15)^{1.5} = 0.1064\ m^3/s$

Triangular ($\theta = 90°$, $\tan 45° = 1$): $Q = (8/15)\, C_d\, \sqrt{(2g)}\, H^{5/2} = (8/15)(0.59)(4.429)\, H^{5/2} = 1.394\, H^{5/2}$

$H^{5/2} = 0.1064/1.394 = 0.0763 \rightarrow H =$ **0.357 m**

Q3 (d)

3 marks

> Explain water hammer in pipes. A pumping plant forces water through a 600 mm main, with friction head 27 m. To reduce power, another main is to be laid alongside so that the two work in parallel over the full length and the friction head drops to 9.6 m. Find the diameter of the new main (same in all other respects).

Water hammer: when water flowing in a long pipe is **suddenly stopped** (rapid valve closure, pump trip), its momentum is destroyed and a **pressure wave of very high intensity** travels back and forth along the pipe at the speed of sound in water (about 1400 m/s), with a hammering noise. The pressure rise for instantaneous closure is $p = \rho V C$ (C = wave speed). It can burst pipes; controlled by slow valve closure, surge tanks, air vessels and relief valves.

Numerical. For the same f and L, $h_f \propto Q^2/d^5$. The total flow Q is unchanged.

Existing pipe alone: $27 \propto Q^2/0.6^5$

Old pipe in parallel: $9.6 \propto Q_1{}^2/0.6^5 \rightarrow Q_1/Q = \sqrt{(9.6/27)} = 0.5963$

New pipe carries $Q_2 = (1 - 0.5963)Q = 0.4037Q$, with $9.6 \propto Q_2{}^2/d^5$

Dividing: $9.6/27 = (0.4037)^2 \times (0.6^5/d^5) \rightarrow d^5 = 0.1630 \times 0.07776/0.3556 = 0.03565$

$d =$ **0.513 m ≈ 513 mm**

Question 4

Q4 (a)

2 marks

> What is the difference between an impulse turbine and a reaction turbine?

Impulse turbine	Reaction turbine
All available head converted to **kinetic energy in a nozzle** before striking the runner	Only part converted to kinetic energy; water enters the runner with **both pressure and kinetic energy**
Pressure stays **atmospheric** throughout the runner	Pressure **changes** as water flows through the runner
Runner need not run full; **casing has no hydraulic function**	Runner passages **run full**; airtight casing needed
No draft tube	**Draft tube** needed
Partial admission (jets on a few buckets)	Full admission all round
High head, low discharge (Pelton wheel)	Medium/low head, high discharge (Francis, Kaplan)

Q4 (b)

3 marks

> A penstock supplies water to a Pelton wheel under a gross head of 500 m; one-third of the gross head is lost in friction. The flow through the nozzle is 2.0 m^3/s and the jet is deflected through 165°. Find the power given to the runner and the hydraulic efficiency. Speed ratio = 0.45, $C_v = 1.0$.

Net head $H = 500 - 500/3 = 333.3$ m

$V_1 = C_v\sqrt{(2gH)} = \sqrt{(2 \times 9.81 \times 333.3)} = 80.87$ m/s; $u = 0.45 \times 80.87 = 36.39$ m/s

$V_{w1} = V_1 = 80.87$ m/s; $V_{r1} = V_1 - u = 44.48$ m/s $= V_{r2}$ (no friction)

Outlet angle $\phi = 180° - 165° = 15°$: $V_{w2} = V_{r2} \cos 15° - u = 42.97 - 36.39 = 6.57$ m/s (backward)

Power to runner $= \rho Q(V_{w1} + V_{w2})u = 1000 \times 2 \times (80.87 + 6.57) \times 36.39 =$ **6.36 MW**

$\eta_h = 2(V_{w1} + V_{w2})u / V_1^2 = 2 \times 87.44 \times 36.39 / 80.87^2 =$ **97.3%**

Question 5

Q5 (a)

3 marks

> Explain the specific speed of a pump. Find the number of identical pumps (800 rpm, specific speed 25, rated capacity 0.16 m^3/s each) required to lift water from a deep well under a total head of 89 m.

Specific speed of a pump (N_s): the speed of a **geometrically similar pump** that would deliver **1 m^3/s against a head of 1 m**: $\mathbf{N_s = N\sqrt{Q} / H^{3/4}}$. It classifies pumps (low N_s: radial flow, high head; high N_s: axial flow, high discharge).

A deep well needs a high head, so the pumps work **in series**, each giving head H_m:

$25 = 800 \times \sqrt{0.16} / H_m^{3/4} \rightarrow H_m^{3/4} = 320/25 = 12.8 \rightarrow H_m = 12.8^{4/3} = 29.94$ m

Number of pumps = 89/29.94 = 2.97 → **3 pumps in series**

Q5 (b)

3 marks

> A centrifugal pump delivers 0.118 m^3/s at 1450 rpm against 25 m head. Impeller diameter 250 mm, outlet width 50 mm, manometric efficiency 75%. Find the vane angle at the outer periphery.

$u_2 = \pi D_2 N/60 = \pi \times 0.25 \times 1450/60 = 18.98$ m/s

$V_{f2} = Q/(\pi D_2 B_2) = 0.118/(\pi \times 0.25 \times 0.05) = 3.005$ m/s

$\eta_{mano} = gH/(V_{w2}u_2) \rightarrow V_{w2} = 9.81 \times 25/(0.75 \times 18.98) = 17.23$ m/s

$\tan \phi = V_{f2}/(u_2 - V_{w2}) = 3.005/(18.98 - 17.23) = 1.717 \rightarrow$ **$\phi = 59.8°$**